

ATISHAYA JAIN

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EDUCATION

B.Tech, Computer Science & Data Science

2023 – 2027

Netaji Subhas University of Technology (NSUT), Delhi · CGPA: 7/10 · JEE Advanced: 10,730 (Gen) · JEE Mains: 13,334 (Gen)

CBSE XII — GD Goenka Public School, Delhi

2023 · 86.2%

RESEARCH

Adaptive Opposition-Based Learning Symbiotic Organisms Search for Robust Portfolio Optimization

First Author · Manuscript in preparation for journal submission

Submission: June 2026

- Designed a novel metaheuristic algorithm fusing opposition-based learning with Symbiotic Organisms Search (SOS) to solve non-convex, high-dimensional portfolio optimization problems; formulated objective functions grounded in stochastic calculus and multivariate probability theory.
- Applied adaptive exploration-exploitation trade-offs with dynamic parameter tuning; validated using Sharpe ratio, Conditional Value-at-Risk (CVaR), and maximum drawdown against mean-variance (Markowitz) and genetic algorithm baselines across multi-asset equity datasets.
- Demonstrated consistent outperformance in risk-adjusted returns; mathematical framework draws on linear algebra (covariance matrix decomposition), numerical optimization, and statistical inference over fat-tailed return distributions.

EXPERIENCE

Data Analyst · PriceWaterhouseCoopers India

June – August 2026 · Gurugram

- Built end-to-end automated data discrepancy finder in Python + Streamlit with an interactive dashboard surfacing near-duplicate records, inconsistent casing, statistical outliers, and unit inconsistencies (kg/lbs/gms → single standard) across large structured datasets.
- Implemented AI-powered semantic category grouping using NLP embeddings to unify fields with different wording but identical meaning (e.g. "SDE" vs "Software Developer"), reducing manual data cleaning effort significantly across client datasets.
- Integrated system with SAP Datasphere for client deployment; identified customers recorded as separate entities across multiple warehouses, enabling data consolidation and resolving duplicate account issues invisible to manual review.

Backend Engineer · Inhouse

May – July 2025 · Remote

- Integrated OCR-based PDF fallback into production chatbot pipeline, resolving content extraction failures that previously affected ~15% of document-processing requests and restoring full system uptime.
- Built automated Markdown-to-Google Docs conversion script, eliminating manual contract formatting and reducing document creation time by ~80%; returned structured document IDs for downstream workflow automation.
- Developed 3+ new REST API endpoints with comprehensive unit test coverage (>95%), triaged and resolved high-severity production bugs within SLA windows throughout the engagement.

PROJECTS

Live BTC Options Trading System

2026 · Live

- Designed and deployed a live covered call options trading system for BTC on Binance, applying options pricing theory (Black-Scholes, delta hedging, implied volatility surface) to structure trades; achieved positive P&L returns from first week of live deployment.
- Engineered real-time signal generation pipeline via Binance WebSocket API for live price feeds with sub-second latency; implemented automated position management, options leg execution, and probabilistic risk controls.
- Built modular Python backtesting framework to validate covered call strategy performance across historical BTC volatility regimes; quantified expected return distributions and tail-risk scenarios using Monte Carlo simulation.

WhatsApp Attendance Bot · 1,000+ Daily Average Users

2024

- Developed WhatsApp chatbot automating attendance extraction from NSUT portal, adopted by 3,000+ unique students with 500 peak concurrent users, eliminating all manual portal navigation.
- Built OCR-powered CAPTCHA bypass and low-latency web scraping pipeline; deployed on Azure Container Instances for auto-scaling using Meta WhatsApp Business API with webhook-based message handling.

TECHNICAL SKILLS

Mathematics: Probability Theory, Stochastic Calculus, Linear Algebra, Real Analysis, Multivariate Calculus, Numerical Optimization, Statistical Inference, Time Series Analysis

Languages: Python (Expert), C++ (Beginner), JavaScript (Beginner)

Quant / Finance: NumPy, pandas, scipy, statsmodels, scikit-learn, QuantConnect, backtrader, vectorbt, pyfolio, Binance API

Databases: PostgreSQL, BigQuery, Redis, MongoDB, CockroachDB

Cloud & Tools: GCP (Cloud Run, Cloud Build, Scheduler), Azure Container Instances, Docker, Git, CI/CD